

Full Length Article

Bidirectional visual-tactile cross-modal generation using latent feature space flow model

Yu Fang^a, Xuehe Zhang^{a,*}, Wenqiang Xu^b, Gangfeng Liu^a, Jie Zhao^a

^a State Key Laboratory of Robotics and System, Harbin Institute of Technology, No. 2, Yikuang Street, Nangang District, Harbin, 150001, Heilongjiang, China

^b Department of Computer Science and Engineering, Shanghai Jiao Tong University, No. 800 Dongchuan Road, Minhang District, Shanghai, 200240, China

ARTICLE INFO

Keywords:

Cross-modal

Visual-tactile data

Deep learning

Flow model

ABSTRACT

Inspired by visual-tactile cross-modal bidirectional mapping of the human brain, this paper introduces a novel approach to bidirectional mapping between visual and tactile data, an area not fully explored in the predominantly unidirectional existing studies. First, we adopt separate Variational AutoEncoder (VAE) models for visual and tactile data. Furthermore, we introduce a conditional flow model built on the VAE latent feature space, enabling cross-modal bidirectional mapping between visual and tactile data using one model. The experimental results show that our method achieves excellent performance in terms of the similarity between the generated data and the original data (Structural Similarity Index (SSIM) of visual data: 0.58, SSIM of tactile data: 0.80), the classification accuracy on generated data (visual data: 91.60%, tactile data: 88.05%), and the zero-shot classification accuracy between generated data and language (visual data: 44.49%, tactile data: 45.03%). To the best of our knowledge, the method proposed in this paper is the first one to utilize a single model to achieve bidirectional mapping between visual and tactile data. Our model and code will be made public after the acceptance of the paper.

1. Introduction

Humans typically rely on visual, tactile, and auditory senses to interact with the world when performing tasks. However, when one of these senses is impaired, it is possible to mitigate this limitation by leveraging information acquired through the other functioning senses. For instance, skilled soldiers can assemble a rifle blindfolded, relying solely on tactile sensations, as they can visualize the entire process by their acute sense of touch. The ability to transfer information across domains and modalities is vital for intelligence (Crick & Koch, 1990; Seymour, Clifford, Logothetis, & Bartels, 2009). The neural pathway in the human brain for visual-tactile bidirectional mapping underpins this capability (Sann & Streri, 2007; Smith & Goodale, 2015). However, current robots cannot interchange information between different senses through cross-modal bidirectional mapping. Our research aims to develop this technology, enhancing robotic perception and functionality, facilitating operations in hidden environments, and improving feedback in visual contexts. This technology also holds significant potential in medical applications, aiding individuals with sensory impairments, as demonstrated in studies such as Liu et al. (2020).

Recently, researchers have widely recognized that the success of cross-modal mapping is attributable to the shared underlying parameters across multiple modalities (Lee, Bollegala, & Luo, 2019; Yuan,

Wang, Dong, & Adelson, 2017). Researchers proposed two datasets to study the cross-modal mapping between visual and tactile modalities: LMT (Strese, Schuwerk, Iepure, & Steinbach, 2016) and Touch2Go (Yang et al., 2022). In the LMT datasets, vibrations represent tactile information, while in the Touch2Go datasets, tactile information is described using tactile images from GelSight (Yuan, Dong, & Adelson, 2017) sensor. The GelSight sensor uses a deformable elastomer piece as the medium of contact, and an embedded camera to capture the deformation of the elastomer surface. These datasets present varied tactile representations and distinct visual coverages. They explored the cross-modal mapping between visual and tactile data using the Generative Adversarial Network (GAN) model (Cai, Zhu, Ban, & Narumi, 2021; Lee et al., 2019; Li, Zhu, Tedrake, & Torralba, 2019; Liu, Wang, Sun, & Fang, 2018). In the GAN model, the generator and discriminator compete with each other. The generator aims to generate samples that are as indistinguishable from original samples as possible, and the discriminator aims to accurately discriminate between generated and original samples. They identified two primary challenges: (1) inherent significant scale discrepancy, arising from the divergent viewing angles between the senses, with vision encompassing the whole object and touch focusing on a tiny fraction of the object; (2) heterogeneous gap, coming from the differing focal points of each sense: vision prioritizes

* Corresponding author.

E-mail address: zhangxuehe@hit.edu.cn (X. Zhang).

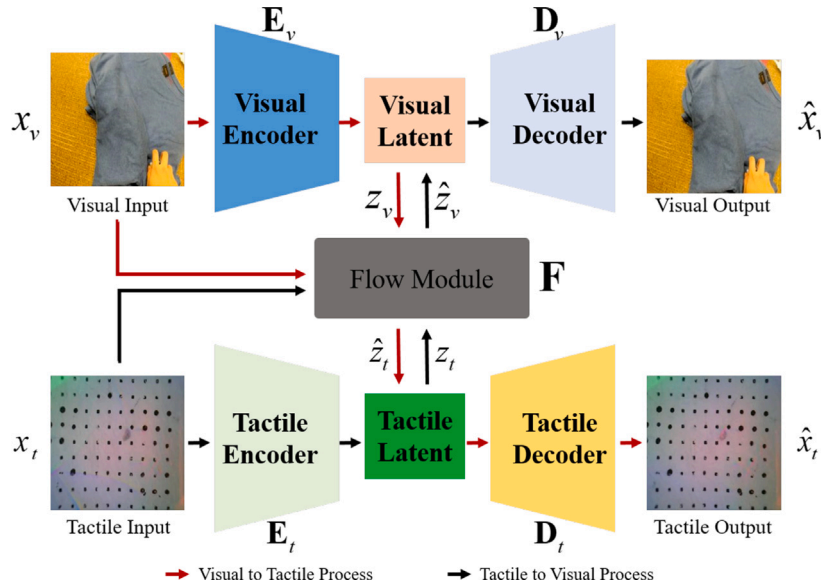


Fig. 1. Overview of the proposed method. It illustrates the interrelation of the components in the proposed method and the data generation process.

aspects like geometry and color, while touch is attuned to surface and texture. To overcome these challenges, they had to develop two GAN models for visual-tactile bidirectional mapping, one for vision-to-tactile mapping and the other for tactile-to-visual mapping.

In our endeavor to bridge the significant scale discrepancy and heterogeneous gap in visual-tactile modalities, we drew inspiration from the mutual mapping mechanisms in the human brain. Research by [Sanchez, Hartmann, Fuscà, Demarchi, and Weisz \(2020\)](#) indicated that early visual and tactile stimuli activate different brain areas, resulting in varied activation signals. After neural processing in the brain, these activation signals can be decoded from one another. Drawing from this understanding, we introduce a bidirectional mapping model for visual-tactile interactions grounded in the latent feature space. We have designed specialized encoders and decoders for visual and tactile data, a bi-directional flow model upon the latent feature space of encoders. We employed these to facilitate the bi-directional mapping of visual and tactile information. A comprehensive overview of the proposed method is illustrated in [Fig. 1](#).

To evaluate our method, we conducted experiments utilizing two visual-tactile cross-modal datasets: Touch2Go and LMT, each offering a unique tactile representation. The experiments demonstrated that our proposed method could generate visual and tactile data that were notably closer to the original data compared to existing methods. Specifically, it surpassed the baseline by 31.14% (tactile) on the Structural Similarity Index (SSIM) metric and reduced the Fréchet Inception Distance (FID) score by 61.65% (visual) and 84.47% (tactile) on the touch2go dataset. In terms of classification, the generated data achieved promising results when assessed using pre-trained visual and tactile classifiers, with scores of 91.60% for visual data and 88.05% for tactile data. Furthermore, the data proved effective in language-based zero-shot classification, demonstrating comparable accuracy to original data — 44.49% (generated data) versus 50.33% (original data) for visual data, and 45.03% (generated data) versus 44.92% (original data) for tactile data, on the Touch2Go dataset. Additionally, we conducted ablation studies to validate the individual components of our model, ensuring their effectiveness and necessity in the proposed method. Using visualization, we clarified the roles of various priors, improving the transparency of the model. Our model is the first to achieve bidirectional cross-modal mapping between visual and tactile domains. In conclusion, we offer the following contributions:

1. **Introduction of a Bidirectional Visual-Tactile Mapping Model:** We have formulated a model capable of facilitating conversions from tactile to visual representations and vice versa.

2. **State-of-the-Art Performance on Visual-Tactile Multimodal Datasets:** We have established new benchmarks through our method, leveraging two visual-tactile multimodal datasets.

2. Related works

Cross-modal research involving visual and tactile domains focuses primarily on generation tasks and multimodal joint learning. Our approach involves an image-to-image translation, so we have briefly reviewed the relevant methods.

Cross-Modal Generation of Visual-Tactile Signals: Visual and tactile signals harbor shared underlying parameters ([Lee et al., 2019](#); [Yuan, Wang, et al., 2017](#)) influencing both modalities, including aspects such as geometry, texture, roughness, and hardness. Such parameters facilitate the cross-modal mapping between the two. A computational module can be employed to extract these shared parameters from one modality and map them to another. Since acquiring tactile cues requires specialized sensors compared to acquiring visual cues, early research centered on generating tactile data from visual input to offset visual perception gaps. The generated tactile data is frequently referred to as pseudo-tactile ([Li et al., 2016](#)). The field primarily features two tactile representations: one using visuotactile sensors for sensation capture and another leveraging vibratory signals. With the visuotactile sensors, [Lee et al. \(2019\)](#) were the first to attempt to realize the generation of tactile texture information from visual data using the conditional GANs approach. [Li et al. \(2019\)](#) utilized the GelSight sensor for tactile data and used GANs to forecast tactile information from visual input. In exploring vibration for tactile characterization, [Ujitoko and Ban \(2018\)](#) automatically generated vibrational tactile signals from texture images using conditional GANs. As the field evolved, researchers started investigating methods for bidirectional transformation between visual and tactile domains. [Cai et al. \(2021\)](#) employed a fusion of visual and tactile data across various scales, combined with perceptual and feature matching losses, to achieve mutual mapping from vibratory to visual signals. While these methods facilitate visual-tactile transitions, their performance in complex scenarios is hindered by the significant scale discrepancy and the heterogeneous gap between the two modalities. Reviewing previous opinions, both the visual-to-tactile and tactile-to-visual mapping processes involve extracting the shared underlying parameters. The current research methods use two independent models to process these mappings, which means that the underlying parameters need to be extracted twice, which may distract the attention of the

model from these parameters. A single model to handle these mappings will allow the model to focus more on the underlying parameters. Additionally, some cross-modal generation methods bypass parameter mapping by constructing a joint representation space (Suzuki & Matsuo, 2022). For instance, the JMVAE model (Suzuki, Nakayama, & Matsuo, 2016) creates a joint multimodal Variational AutoEncoder (VAE) connecting all modes with the joint representation. It captures high-level concepts and enables multimodal sampling through joint Kullback-Leibler (KL) divergence training. JNF (Senellart, Chadebec, & Allassonnière, 2023) further improves this with the Normalizing Flow and correlation analysis for arbitrary multimodal extensions. While these methods are theoretically promising, they have yet to be validated on complex cross-modal data generation tasks, such as visual-tactile cross-modal generation.

Multimodal Learning for Visual-Tactile: This field seeks to identify shared information and link two modalities, improving performance in downstream tasks. Common downstream tasks include classification, reconstruction, and robot manipulation. For classification: Gao, Hendricks, Kuchenbecker, and Darrell (2016) merged visual and tactile signals from the PHAC-2 (Chu et al., 2015) dataset, enhancing touch-based surface classification accuracy. The research conducted by Yuan, Wang, et al. (2017) integrated vision, depth, and touch images to classify fabrics. Through a comparative analysis of physical feature embeddings across different modalities, they identified shared information between these modalities. They also showed that combined multimodal input outperformed single-modality input. For reconstruction: Xu et al. (2023) combined visual and tactile data to reconstruct rigid and non-rigid objects. For manipulation: Cui, Wang, Wei, Hu, and Wang (2020) proposed a novel visual-tactile fusion learning method based on the self-attention mechanism to complete the robot grasping. Yu et al. (2023) presented a visual-tactile approach to manipulate the deformable linear object with reinforcement learning. These methods underscore the intricate link between vision and tactile sensation, allowing researchers to enhance robotic perception. The cross-modal generation approach offers a way to achieve multimodal learning when some modalities are absent, using an algorithm to integrate generated missing modalities with existing modalities to facilitate multimodal learning. However, an essential prerequisite is ensuring the high quality of the generated data.

Image-to-Image Translation: This task focuses on converting an input image from one domain to a realistic picture in another. It requires working exclusively within the image domain to ensure geometrical alignment between input and output. Contrary to the image domain, the visual-tactile domain is the weak geometrical alignments (Cai et al., 2021), given that data from both modalities are perceived at varying scales. Image-to-image transformation does not need a one-to-one correspondence, but visual-tactile mappings usually do. These differences make it challenging to directly apply image-to-image translation techniques to visual-tactile mapping, prompting a unique exploration of cross-modal algorithms. Early image-to-image methods mainly used GAN. For example, Zhu et al. (2017) used GANs for day-night transformations, Isola, Zhu, Zhou, and Efros (2017) for contour-based image generation, and Zhu, Park, Isola, and Efros (2017) for scene transformations. Owing to the advancements in generative methods, such as the Diffusion and Normalizing Flow approach have recently improved the performance of image-to-image tasks. Ardizzone et al. (2021) used conditional reversible neural networks for tasks like day-to-night translation and diverse image colorization. Saharia et al. (2022) developed an image-to-image diffusion model for applications like colorization and inpainting. In addition to GAN, Diffusion, and Normalizing Flow, the energy-based method has also been used for image-to-image translation (Zhao, Xie, & Li, 2020).

3. Vision-touch bidirectional flow

The visual domain is denoted as D_v , and the tactile domain is denoted as D_t . We aim to create an algorithm for mutual mappings

between the two domains, represented as $D_t \leftrightarrow D_v$. We use devices like cameras to capture the visual signal $x_v \in \mathbb{R}^{3 \times H_v \times W_v}$ from D_v . At the same time, using tactile sensors, we gather signals from D_t , which are represented as vibrations or tactile images. For our method, tactile signals are depicted as images, represented by $x_t \in \mathbb{R}^{3 \times H_t \times W_t}$. The substantial inherent disparities between the two domains, coupled with varying signal representations from different devices, give rise to two primary challenges: a notable scale discrepancy and a heterogeneous gap between the modalities. These issues make bidirectional mapping of modalities between vision and touch very difficult. The objective of the proposed method is to address these challenges.

To address the challenges, we propose a method for constructing bi-directional mapping models in latent feature space inspired by the mechanism of visual-tactile cross-modal mapping in the human brain. First, we define two encoders: E_v for visual and E_t for tactile inputs, mapping them to their respective latent feature spaces. Then, we introduce a flow module F , that enables bidirectional mapping between these latent feature spaces. A visual image x_v or tactile image x_t , when processed by E_v or E_t , respectively, yields a low-dimensional embedding z_v^e or z_t^e , each parameterized as a Gaussian distribution: $\mathcal{N}(\mu(z_v^e), \sigma^2(z_v^e))$ or $\mathcal{N}(\mu(z_t^e), \sigma^2(z_t^e))$. We draw samples z_v for visual and z_t for tactile from these distributions. The flow module then converts these samples into their corresponding tactile embeddings $\hat{z}_t$ or visual embeddings $\hat{z}_v$. Subsequently, D_t for tactile and D_v for visual produces signals $\hat{x}_t$ and $\hat{x}_v$ from these embeddings, respectively. The complete workflow of the algorithm is illustrated in Fig. 1. We will delve deeper into each component of our algorithm in the following sections.

3.1. Tactile and visual variational AutoEncoder

Our bidirectional mapping module is built on the visual and tactile encoder latent feature space. Unlike directly establishing a bidirectional mapping model in the original visual representation and tactile representation space, utilizing latent feature space offers the following advantages: 1. The disparity between visual and tactile signals in the original space is substantial. However, in the latent feature space, as all samples are constrained to the family of normal distributions, the differences between images can be reduced. 2. It provides a concise solution for high-resolution images without significantly increasing computational overhead.

Building on the approach in Rombach, Blattmann, Lorenz, Esser, and Ommer (2022), we utilize a variational autoencoder to map the original data space to a latent space. The vision encoder E_v produces a latent representation z_v that retains its two-dimensional structure instead of being flattened. Specifically, $z_v \in \mathbb{R}^{C_v \times \frac{H_v}{m} \times \frac{W_v}{m}}$, with m as the downsampling factor, and in our setting, $m = 8$. This method is similarly applied to tactile images. Figs. 2(a) and 2(b) show the visual and tactile VAE structures, respectively.

In the training phase, beyond the reconstruction loss $\mathcal{L}_1(\hat{x}, x) = \|x - \hat{x}\|_1$, KL divergence $KL(z^e \| I) = 0.5(\mu(z^e) + \sigma(z^e) - 2 \log \sigma(z^e) - 1)$, we integrate a perceptual loss (Zhang, Isola, Efros, Shechtman, & Wang, 2018): $\mathcal{L}_{\text{percep}}(\hat{x}, x) = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \|y_{hw}^l - \hat{y}_{hw}^l\|_2^2$, where $y_{hw}^l, \hat{y}_{hw}^l$ are features extracted from $x, \hat{x}$ in the l th layer of a deep neural network (e.g. VGG network (Simonyan & Zisserman, 2015)) respectively and unit-normalized in the channel dimension. The perceptual loss counters the blurriness arising from exclusive reliance on pixel-space loss. In total, our VAE loss $\mathcal{L}_{\text{AELoss}}$ is composed of three elements:

$$\mathcal{L}_{\text{AELoss}} = \alpha * \mathcal{L}_1(\hat{x}_i, x_i) + \beta * KL(z_i^e \| I) + \lambda * \mathcal{L}_{\text{percep}}(\hat{x}_i, x_i) \quad (1)$$

Where $\alpha = 1.0$, $\beta = 1e - 6$, $\lambda = 1.0$, and i represents the visual or tactile modality.

3.2. Flow module

We employ the flow model F as a computational module that connects two latent feature spaces. The significant characteristic of the

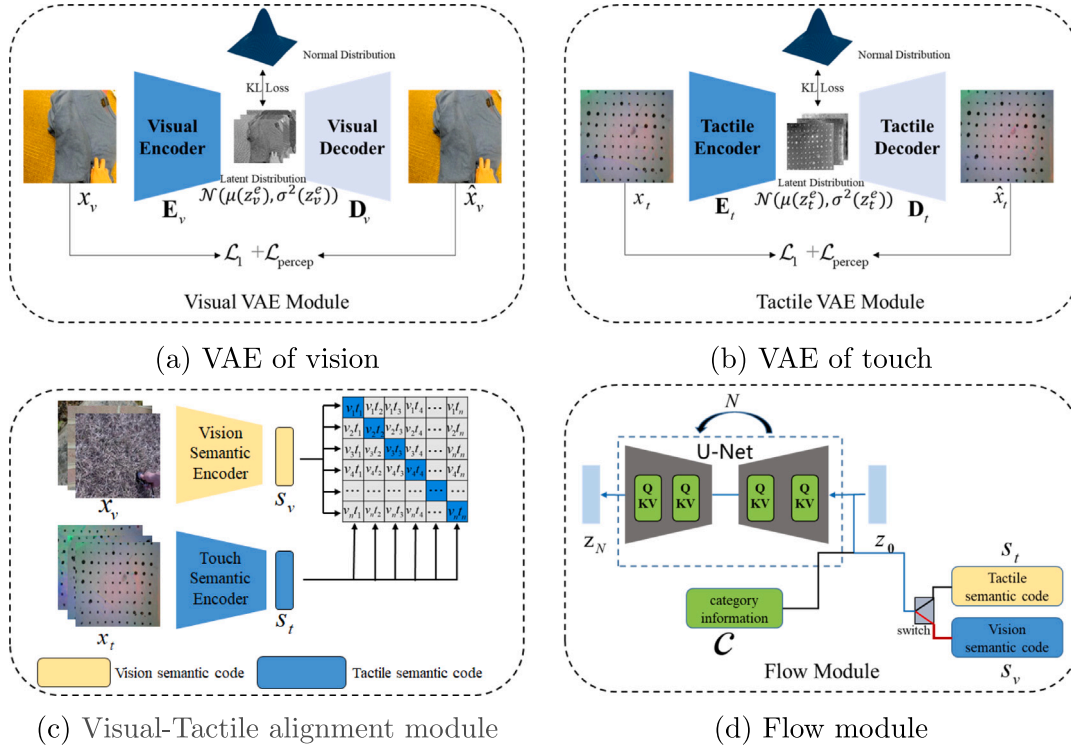


Fig. 2. The components of the proposed method (a) and (b), respectively, depict the visual VAE model and the tactile VAE model utilized for compressing visual and tactile data. (c) illustrates the process of aligning visual and tactile data; (d) shows details of the processing module employed for bidirectional mapping in visual and tactile latent feature spaces.

flow model is its ability to establish a bidirectional mapping between two distributions. This characteristic enables the implementation of two mapping processes through a single flow model. The two distributions are defined as visual and tactile latent feature space in our task. The forward process of the flow model is the mapping from the tactile to the visual latent feature space, denoted as $z_v = \mathbf{F}(z_t)$, and the reverse process is the mapping from the visual to the tactile latent feature space, represented as $z_t = \mathbf{F}^{-1}(z_v)$.

Prior: We can implement cross-modal mapping because they all contain underlying parameters. If we provide more information about them, it will be beneficial to alleviate the difficulty of mapping. Therefore, we introduce two kinds of priors to reduce the difficulty of mapping.

First, we introduce some information that only depends on the object itself. This information type is closely associated with the underlying parameters and spans various modalities. The representative of this is category information.

Secondly, we aim to extract information about the underlying parameters within a single modality. Since the underlying parameters are never directly observed, we can only approximate them through learning from various modalities, but it is impossible to get them. The underlying parameters are properties shared by multiple modalities, meaning aligned information in the different modalities is most closely related to the underlying parameters.

Thus, adopting multi-modal self-supervised learning is a viable learning approach to obtaining aligned information. This method obtains the aligned information between modalities by aligning each modality. Consequently, we incorporate semantic code derived from visual-tactile alignment pre-trained as prior information. Building on the success of CLIP (Radford et al., 2021), we adopt its trained visual encoder to extract visual semantic code. Then, we define a tactile encoder with the same structure as the visual encoder to extract tactile

semantic code. Fig. 2(c) illustrates the process of getting semantic code. Like CLIP, we employ the Info Noise Contrastive Estimation (InfoNCE) loss for supervision:

$$\mathcal{L}_{InfoNCE} = -\log \frac{\exp(f(s_v^i)^T f(s_v^j)/\tau)}{\sum_{j=1}^n \exp(f(s_v^i)^T f(s_v^j)/\tau)} \quad (2)$$

Where f is l_2 norm, τ is temperature parameter, s_v is semantic code from vision, s_t is semantic code from touch.

Bidirectional Flow: We choose the RectifiedFlow (Liu, Gong, & Liu, 2022) as a bridge for the mutual mapping of the two modalities, which is a simple approach to learning ordinary differential equation (ODE) models to transport between two empirically observed distributions. We formulate a differential equation between two modalities as $z_{target} = z_{source} + \int_0^1 \omega(z_\eta, \eta) d\eta$, where $z_\eta = \eta z_{target} + (1 - \eta) z_{source}$, and $z_{source}, z_{target} \in \{z_v, z_t\}, z_{target} \neq z_{source}, \eta \in [0, 1]$. Using a neural network, it learns the velocity $\omega(z_\eta, \eta)$ between visual and tactile modalities. Subsequently, it employs the Euler method to solve the differential equations, establishing a mutual mapping between these modalities.

We integrate priors into our network model to estimate the velocity $\omega(z_\eta, \eta, c, s_i)$. Here, c denotes category information, $s_i \in \{s_t, s_v\}$, where s_v is visual semantic code, and s_t is tactile semantic code. In the original RectifiedFlow, inverse mapping between distributions is realized by inverting the sample flow direction. Specifically, after the neural network training is completed, $\omega(z_\eta, \eta, c, s_{source})$ can be estimated, the forward process is: $z_{source} \rightarrow z_{target} : z_{\eta+\Delta\eta} = z_\eta + \omega(z_\eta, \eta, c, s_{source})\Delta\eta$ and the reverse is: $z_{target} \rightarrow z_{source} : z_\eta = z_{\eta+\Delta\eta} - \omega(z_{\eta+\Delta\eta}, c, s_{source})\Delta\eta$. So, a contradiction arises: we need to use s_{source} to estimate z_{source} , while we only have s_{target} from z_{target} . To address this, we suggest employing a network that concurrently calculates velocities $\omega(z_\eta+\Delta, c, s_{source})$ and $\omega(z_\eta+\Delta, c, s_{target})$. This method enables concurrent estimation velocities of both processes during training, overcoming the single mapping process constraint. Our approach utilizes semantic code to dictate the

Table 1
Results on the Touch2Go dataset.

		SSIM↑	PSNR↑	FID↓	CA↑	FLOPs↓	IT(ms)↓
T2V	VTM	0.47	17.23	106.83	64.05%	99.36G	34.35
	Pix2Pix	0.48	16.54	115.99	24.13%	9.03G	7.47
	JNF	0.55	20.16	312.02	18.73%	9.99G	115.06
	Ours	0.58	22.33	40.96	91.60%	459.43G	200.67
V2T	VTM	0.60	22.00	27.76	62.50%	99.36G	34.35
	Pix2Pix	0.50	18.15	119.42	26.11%	9.03G	7.47
	JNF	0.61	19.44	268.87	10.98%	9.99G	115.06
	Ours	0.80	25.71	4.31	88.05%	459.43G	200.67

direction of flow, a refinement that negates the necessity of reversing the flow when mapping between two bidirectional distributions. Fig. 2(d) shows the complete flow module.

For tactile-to-visual and visual-to-tactile mappings, we define transitional states as $z_{\eta}^{vt} = \eta z_v + (1 - \eta) z_t$ and $z_{\eta}^{tv} = \eta z_t + (1 - \eta) z_v$. The overall training loss for the entire process is given by:

$$\mathcal{L} = \min_{\omega} \int_0^1 \mathbb{E} \left[\|(z_v - z_t) - \omega(z_{\eta}^{vt}, \eta, c, s_t)\| \right] + \min_{\omega} \int_0^1 \mathbb{E} \left[\|(z_t - z_v) - \omega(z_{\eta}^{tv}, \eta, c, s_v)\| \right] \quad (3)$$

4. Experiments

Our experiments seek to answer the following: 1. Can our method be generalized across multiple scenarios? 2. How do data generated from our method differ from original data in various categories? 3. How well do data generated perform on other tasks? These questions evaluate the consistency of the data generated by our method with original data from different views.

4.1. Experiments settings

Baseline: To evaluate the performance of the proposed method, we compare the results from our model, VTCM (Cai et al., 2021), Pix2Pix (Isola et al., 2017) and JNF (Senellart et al., 2023). VTCM is based on a conditional GAN model with residual fusion and contrasts features across different levels, which performed well in LMT datasets. The Pix2Pix model was used in the previous work on robotic cross-modal vision-to-touch generation (Lee et al., 2019). JNF is a joint VAE model that combines correlation analysis and the Normalizing Flow.

Evaluation Metrics: Based on methods from Cai et al. (2021), Lee et al. (2019), we first trained two Densenet121 classifiers using original data. We then used these classifiers to assess the generated visual and tactile data in terms of classification accuracy (CA). Additionally, we employed the SSIM, Peak Signal-to-Noise Ratio (PSNR), and FID to measure differences between generated data and original data. We also evaluated the FLOPs and inference time (IT) of different models. SSIM assesses image quality by considering the brightness, contrast, and structure of images, which range from 0 to 1, with higher values denoting more image similarity. FID uses the Inception-v3 network (Szegedy, Vanhoucke, Ioffe, Shlens, & Wojna, 2016) to measure the Frechet distance between generated data and original data, commonly used to evaluate the performance of generated images.

Datasets: Our experiments were conducted on two datasets: LMT and Touch2Go. The LMT dataset has 108 categories of materials, including 20 sets of RGB surface images and acceleration-based tactile signals. For fairness, we selected pre-processed data in Cai et al. (2021) as the experimental dataset, which included 9-class/type subsets. After data augmentation, 1000 samples were obtained for each category, resulting in a weak pairing between vision and touch. The Touch2Go dataset gathers visual and tactile data from a first-person human viewpoint. As objects are detected using tactile sensors, simultaneous visual data is captured from the same human perspective. This dataset encompasses object categories across diverse indoor and

outdoor environments, with over 3971 instances, making it intricate. The LMT dataset and the Touch2Go dataset are two distinct types of datasets, each characterized by unique haptic representations and collected under varying conditions. The LMT dataset specifically focuses on different types of material visual and haptic data within a sample scene, representing haptic signals via frequencies. In contrast, the Touch2Go dataset encompasses a broader range of scenes and utilizes images to represent haptic signals. The primary objective of conducting experiments with these two diverse datasets is to more effectively and comprehensively assess the performance of our proposed method.

Training Process: Initially, we trained two VAE models to condense visual and tactile data into a low-dimensional space. We employed the Adam optimizer, setting a learning rate at $4.5e-6$, a batch size of 6. Next, we established a tactile-visual alignment module. During this phase, we froze the weights of the visual encoder during training and only updated the weights of the tactile encoder. We used the AdamW optimizer and set a learning rate at $5e-4$, a batch size of 384. Lastly, we trained the flow model on the latent feature space of visual and tactile modalities. We employed UNet to estimate velocity and used the Euler integral to generate data. We used the AdamW optimizer, setting a learning rate at $2e-5$, with a weight decay of 0.1.

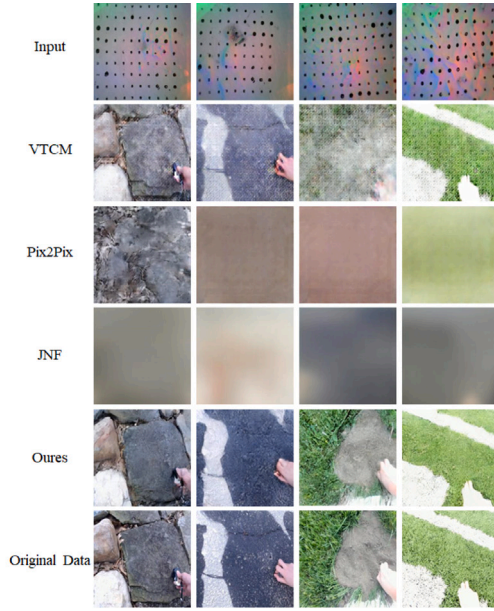
Note: Since the LMT dataset only contains simple surfaces, we did not use semantic code to control the generation direction in flow. We only integrated the category prior, resulting in a simplified version of our method.

4.2. Experiments results

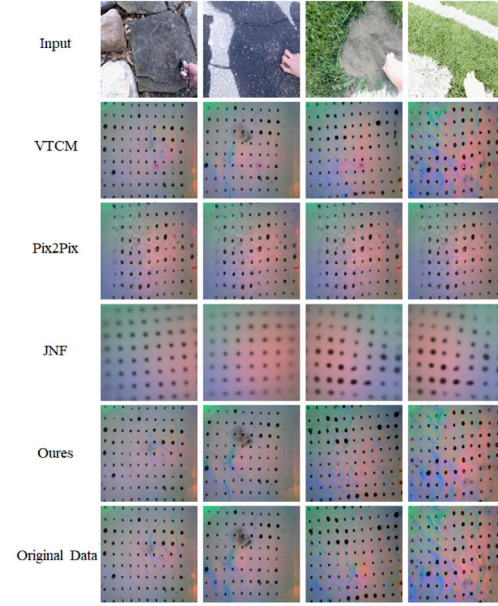
Our method outperformed baselines, yielding impressive results on both datasets. For the Touch2Go dataset, our generated visual data had a classification accuracy of 91.60%; generated tactile data achieved 88.05% and an FID of 4.31. For the LMT dataset, our generated visual data boasted a 99.87% classification accuracy; tactile data gained 100% and an FID of 2.41. For detailed results, please take a look at Table 1 (Touch2Go) and Table 2 (LMT). Fig. 3 visually illustrates the generated results on Touch2Go and LMT datasets.

In both datasets, the quality of generated tactile data surpassed that of vision. This is attributed to the intrinsic properties of vision and touch, as visual information encompasses a broader range than tactile signals, making it more challenging to generate vision than touch. However, a notable observation surfaced during the classification evaluation of the generated signals on the Touch2Go dataset. Even with the superior quality of tactile signals, classifiers performed markedly better on visual signals. This trend is not exclusive to our method, it is also evident in baselines. The inherent similarity in tactile signals impedes classifiers from distinguishing between categories, while visual signals represent abstract category details more effectively. This insight enhances our understanding of intricacies in cross-modal research, offering valuable guidance for subsequent studies.

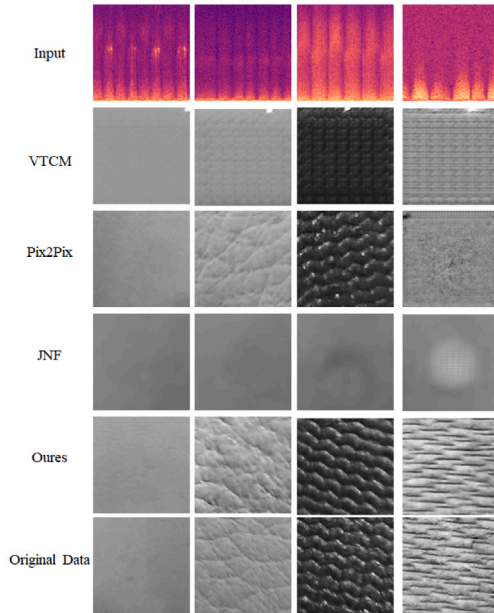
We have visualized the procedure to help us understand the generation process of our method, shown in Fig. 4. The results indicate the encoder adeptly captures prior information from visual and tactile modalities. Even without flow module processing, using the tactile decoder to decode the signal compressed by the visual encoder, the



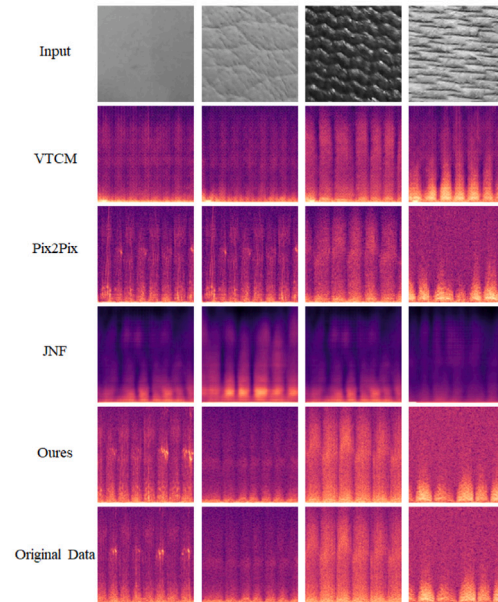
(a) Generation sample of touch2vision on the Touch2Go



(b) Generation sample of vision2touch on the Touch2Go



(c) Generation sample of touch2vision on the LMT



(d) Generation sample of vision2touch on the LMT

Fig. 3. Examples of cross-modal generation. (a) is touch $\rightarrow$ vision on the Touch2Go, and (b) is vision $\rightarrow$ touch on the Touch2Go, (c) is touch $\rightarrow$ vision on the LMT, (d) is vision $\rightarrow$ touch on the LMT. The first row shows the input information, the second row represents the result generated by the VTCM, the third row represents the generated result of Pix2Pix, the fourth row represents the generated result of JNF, the fifth row represents the generated result of ours, and the last row represents the original data.

Table 2
Results on the LMT dataset.

	method	SSIM $\uparrow$	PSNR $\uparrow$	FID $\downarrow$	CA $\uparrow$	FLOPs $\downarrow$	IT $\downarrow$
T2V	VTCM	0.40	16.68	224.87	20.47%	99.36G	34.35
	Pix2Pix	0.33	15.97	184.66	70.70%	9.03G	7.47
	JNF	0.44	15.76	264.85	10.27%	9.99G	115.06
	Ours	0.44	15.73	74.55	99.87%	459.43G	200.67
V2T	VTCM	0.49	20.01	57.27	88.12%	99.36G	34.35
	Pix2Pix	0.53	19.68	70.55	89.90%	9.03G	7.47
	JNF	0.69	20.68	183.01	9.87%	9.99G	115.06
	Ours	0.56	20.58	2.41	100.00%	459.43G	200.67

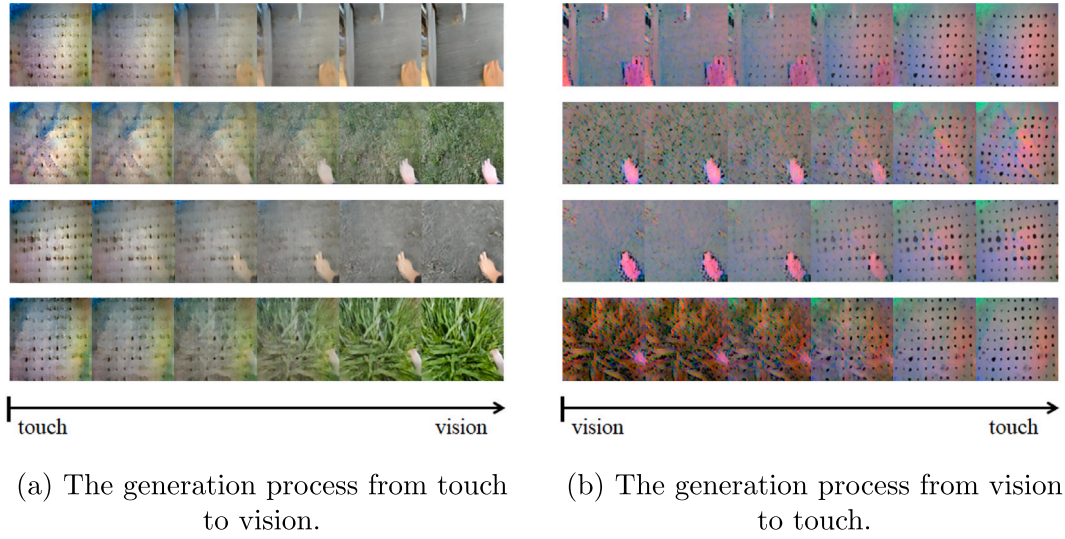


Fig. 4. The examples of the generation process. (a) illustrates the tactile data generation, while (b) depicts the corresponding visual data process.

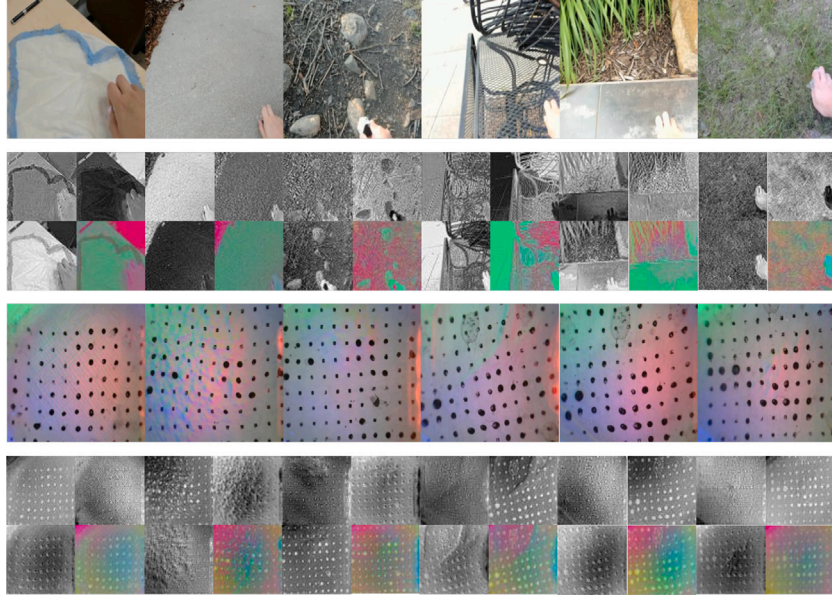


Fig. 5. Examples of visualizing latent feature space. The first row is the visual original image, the second is the visualized image of the visual latent feature space, the third is the tactile original image, and the last is the visualized image of the tactile latent feature space.

obtained signal also exhibits obvious stylistic characteristics of the tactile signal. As the number of iterations increases (the parameter η rises), the method progressively sharpens intricate contours and details. This observation is consistent when converting signals from tactile to visual.

To grasp the characteristics of latent feature space, we visualized it. Fig. 5 depicts the visualized samples from the latent feature space. The visual images show that the pieces retain certain shape features within the latent feature space, and local contours can still be distinguished. We investigated if the VAE model could minimize the distance between two distributions. We used Maximum Mean Discrepancy (MMD) to measure the distance between tactile and visual data, a common statistical metric for measuring the distance between two distributions. We computed the MMD with the Laplacian kernel between the original visual and tactile spaces, obtaining values of 0.7415 and 0.4379 for the visual latent and tactile latent feature spaces, respectively. This indicates that our method, utilizing latent feature space encoding, reduces the distance between the two distributions, simplifying the mapping process.

4.3. Ablation

We conducted four experiments: no prior condition, adding category prior, adding category prior with direction-agnostic semantic code, and our method. In the direction-agnostic method, we randomly selected two types of semantic code as a prior condition. For each metric, we evaluated over 4 random seeds and reported mean scores and 95% confidence intervals. Our experimental results showed that proposed priors can improve the final performance of the algorithm. We found that the method of adding category prior with direction-agnostic semantic code had little impact on performance versus only adding category prior. However, the method using semantic coding to control the direction of flow can greatly improve the final performance of the algorithm. We also found greater quality improvement in generating tactile versus visual data with our method.

As shown in Table 3, the results from our experiments demonstrate that incorporating category information enhances classification accuracy in visual and tactile generation tasks, with an average improvement of about 4%. We compared the generated data, as shown in

Table 3

Results on the Touch2Go dataset for different conditional approaches with means and 95% confidence intervals calculated over 4 seeds.

	method	SSIM↑	PSNR↑	FID↓	CA↑
T2V	no condition	0.561 ±0.003	21.84 ±0.13	55.30 ±2.07	83.73% ±1.94%
	only category condition	0.565 ±0.001	21.97 ±0.13	49.15 ±3.42	88.43% ±0.62%
	category+semantic (direction-agnostic)	0.565 ±0.004	21.96 ±0.08	51.21 ±0.52	88.63% ±0.35%
	category+semantic (ours)	0.576 ±0.001	22.39 ±0.06	41.92 ±2.92	91.07% ±0.89%
V2T	no condition	0.730 ±0.004	23.78 ±0.14	8.88 ±1.32	74.71% ±0.40%
	only category condition	0.739 ±0.005	23.98 ±0.15	7.64 ±0.71	78.26% ±0.63%
	category+semantic (direction-agnostic)	0.741 ±0.005	24.01 ±0.13	7.52 ±0.89	78.66% ±1.30%
	category+semantic (ours)	0.796 ±0.002	25.67 ±0.10	4.49 ±0.26	88.10% ±0.59%

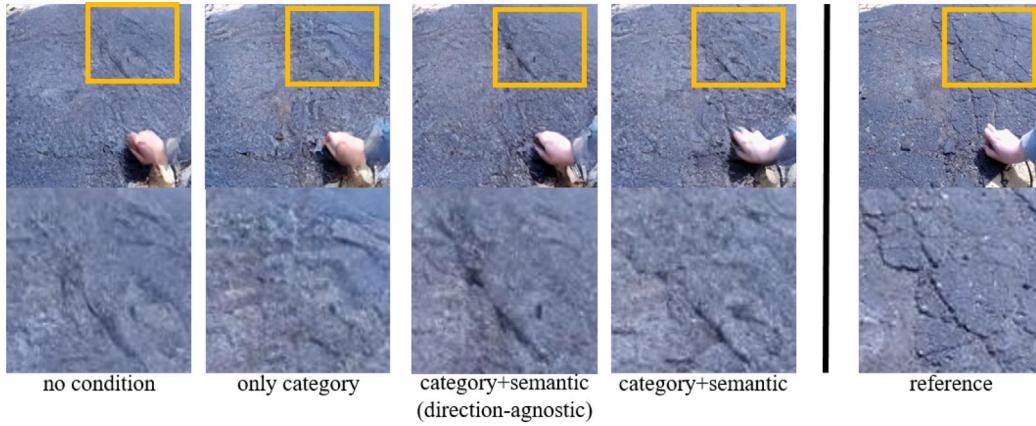
**Fig. 6.** Examples of vision generated under different priors. The first row is the entire image, and the second is an enlarged image from the cropped area.

Fig. 6, and found that the added semantic code can enrich the details of the generated data. Additionally, we observed that decoupling the flow direction has the greatest impact on the performance of generated data. For generated tactile data, the CA can be improved by 12.00% and the FID can be reduced by 40.29%. Overall, these results emphasize the crucial role of category, semantic code, and decoupling the flow direction in enhancing both the quality and accuracy of the model generated for visual and tactile tasks, showcasing the potential of our method.

Using the Touch2Go dataset, we verified the ability of the semantic code to control the flow direction. For each method, we evaluated 4 random seeds and reported the mean scores and 95% confidence intervals. Expressly, when generating tactile signals from visual inputs, we conditioned on the semantic code of the two inputs and compared the output, finding that the sum of their velocities was almost zero. This observation suggests that the semantic code can precisely control the opposing flow directions. We assume we have the corresponding vision semantic code to produce a visual signal and subsequently invert the output velocity to create a visual sample. It is important to note that this process is not a general practice. The primary objective here was to validate the semantic code's ability to effectively control the flow direction, a critical aspect of our methodology. We compared the quality of these generated data with those produced using our method to delineate the efficacy of our approach. Experimental findings demonstrate that both methods produce data with similar performance across various evaluation metrics. Please take a look at Table 4 for detailed results.

To conduct a comprehensive investigation into the role of category prior to our generation process, we analyzed the category embeddings. We visualized the outcomes generated under the influence of varying

category information. Initially, we scrutinized the category embeddings, which unveiled a rich diversity in the representations within the latent feature space. It was noted that the cosine distances between category embeddings hover around 1, indicating a near orthogonality in the high-dimensional space. This characteristic indicates that category embeddings can effectively differentiate between various object classes. Concurrently, it suggests that altering category embeddings can influence the style of the generated data. Therefore, we allocated different category embeddings to a single sample to observe the impact on the outcomes generated. The experimental results demonstrate that altering category embeddings enables the generation of images with diverse styles. Fig. 7 illustrates the variations in style when utilizing different category embeddings. Furthermore, the experimental results reveal that our method goes beyond mere data fitting; it adeptly identifies and leverages the characteristic styles of various categories by utilizing category information.

To assess the impact of varying β values, we conducted comparative experiments within the $\mathcal{L}_{AELoss}$ framework. We selected three distinct parameter sets: 1e-3, 1e-5, and 1e-6. Our evaluations focused on four key metrics: SSIM, PSNR, FID, and CA. The results indicate that different β values do not consistently affect all indicators in the same manner, and the overall variation among them is minimal. This suggests that the β coefficient has a limited impact on the outcomes. For a comprehensive view of these results, please refer to Table 5.

4.4. Data generation for different categories

To assess the generative quality of our method in generating vision across various categories, we conducted a detailed analysis of the classification results derived from the Touch2Go dataset. We chose to utilize

Table 4

Results on the Touch2Go dataset for different methods with means and standard deviations calculated over 4 seeds.

	method	SSIM↑	PSNR↑	FID↓	CA↑
T2V	semantic code (vision)	0.553	21.78	40.29	92.04%
		± 0.002	± 0.06	± 2.06	$\pm 0.94\%$
	semantic code (tactile)	0.576	22.39	41.92	91.07%
		± 0.001	± 0.06	± 2.92	$\pm 0.89\%$
V2T	semantic code (vision)	0.796	25.67	4.49	88.10%
		± 0.002	± 0.10	± 0.26	$\pm 0.59\%$
	semantic code (tactile)	0.785	25.32	3.58	87.31%
		± 0.003	± 0.12	± 0.14	$\pm 0.32\%$

**Fig. 7.** Examples of image generation in different styles. By inputting different category embeddings, images of different styles can be generated.**Table 5**The impact of different β on the results.

	β	SSIM↑	PSNR↑	FID↓	CA↑
T2V	$\beta = 1e-6$	0.58	22.42	39.81	91.50%
	$\beta = 1e-5$	0.58	22.48	45.78	92.83%
	$\beta = 1e-3$	0.59	22.57	39.41	91.74%
V2T	$\beta = 1e-6$	0.80	25.68	4.44	88.56%
	$\beta = 1e-5$	0.80	25.74	6.33	87.01%
	$\beta = 1e-3$	0.81	26.45	5.02	89.33%

the F1 score instead of the previously used accuracy rate, as it offers a more balanced measure by harmonizing precision and recall, thereby providing a comprehensive assessment of the classification model. Following the previous settings, we evaluated 4 random seeds and reported the mean scores and 95% confidence intervals. We computed the confusion matrix and F1 score for every category. The gravel class has the lowest F1 score at 0.640 ± 0.159 . Upon inspecting the confusion matrix, the accuracy of our method was 99.58%, but its precision dropped to 65.00%. Most instances predicted as gravel by the model were soil. This inconsistency could stem from the common overlap of these two

classes in related scenes. The Nat-Fabric(nature fabric) ranks second lowest, with an F1 score of 0.763 ± 0.047 . This low score resulted from a diminished precision. Please take a look at Table 6 for more details.

We also calculated the confusion matrix and F1 score of the tactile classifier. The sand class had the lowest F1 score at 0.713 ± 0.119 . The model misclassified some sand instances as Syn-Fabric(synthetic fabric), leading to a 60% recall rate. This error notably impacted the overall performance of the sand class. The metal followed with an F1 score of 0.732 ± 0.027 . A deeper look showed the low score was primarily due to a recall of just 58.06%. We noted a frequent

Table 6

F1 score for different categories on generated data with means and 95% confidence intervals calculated over 4 seeds.

Class	Accuracy↑		Precision↑		Recall↑		F1 score↑	
	Vision	Tactile	Vision	Tactile	Vision	Tactile	Vision	Tactile
Concrete	97.34%	96.20%	86.92%	83.92%	95.30%	90.04%	0.909	0.869
	±0.39%	±0.20%	±1.69%	±1.23%	±1.20%	±0.76%	±0.013	±0.006
Plastic	98.34%	98.21%	82.74%	89.91%	83.54%	71.53%	0.831	0.797
	±0.22%	±0.10%	±2.83%	±2.27%	±2.43%	±1.51%	±0.024	±0.011
Glass	99.73%	99.67%	88.73%	85.50%	81.58%	77.63%	0.850	0.813
	±0.15%	±0.13%	±10.41%	±6.36%	±7.6%	±9.97%	±0.083	±0.076
Wood	97.54%	93.48%	92.04%	75.96%	94.78%	94.28%	0.934	0.841
	±0.36%	±0.46%	±1.54%	±1.55%	±1.21%	±0.93%	±0.009	±0.010
Metal	97.86%	96.18%	94.44%	99.09%	81.00%	58.06%	0.872	0.732
	±0.19%	±0.29%	±2.22%	±0.80%	±1.66%	±3.58%	±0.011	±0.027
Brick	99.76%	98.57%	97.21%	95.05%	98.73%	79.13%	0.980	0.864
	±0.14%	±0.16%	±2.63%	±2.33%	±0.55%	±1.39%	±0.012	±0.015
Tile	98.84%	98.95%	93.42%	81.42%	66.87%	85.32%	0.779	0.833
	±0.19%	±0.17%	±3.73%	±5.46%	±6.79%	±1.63%	±0.045	±0.021
Leather	99.83%	99.70%	97.96%	93.63%	95.05%	94.31%	0.965	0.940
	±0.11%	±0.08%	±2.27%	±2.51%	±2.88%	±2.36%	±0.024	±0.016
Syn-Fabric	96.69%	98.67%	93.39%	97.29%	91.94%	96.82%	0.927	0.971
	±0.35%	±0.30%	±0.98%	±0.46%	±0.72%	±0.97%	±0.008	±0.007
Nat-Fabric	99.88%	100.00%	61.72%	100.00%	100.00%	100.00%	0.763	1.00
	±0.03%	±0.00%	±6.20%	±0.00%	±0.00%	±0.00%	±0.047	±0.000
Ruber	99.98%	99.99%	94.51%	100.00%	100.00%	97.92%	0.971	0.989
	±0.04%	±0.02%	±10.97%	±0.00%	±0.00%	±6.63%	±0.059	±0.035
Paper	99.25%	99.24%	79.09%	86.39%	92.19%	79.95%	0.851	0.830
	±0.09%	±0.15%	±3.45%	±6.08%	±3.95%	±3.42%	±0.017	±0.029
Tree	99.90%	99.81%	91.92%	96.10%	98.84%	85.47%	0.952	0.905
	±0.04%	±0.09%	±2.94%	±5.28%	±2.14%	±3.54%	±0.017	±0.043
Grass	99.55%	99.31%	99.50%	90.08%	91.76%	97.57%	0.955	0.937
	±0.09%	±0.18%	±1.12%	±2.64%	±0.91%	±0.93%	±0.008	±0.016
Soil	99.50%	99.59%	82.98%	88.02%	85.16%	85.55%	0.840	0.867
	±0.15%	±0.12%	±6.01%	±4.47%	±3.21%	±5.87%	±0.043	±0.038
Rock	99.05%	99.11%	90.37%	91.33%	95.25%	95.06%	0.927	0.932
	±0.28%	±0.16%	±2.70%	±0.78%	±1.88%	±3.00%	±0.021	±0.014
Gravel	99.58%	99.88%	65.00%	91.82%	63.46%	89.00%	0.640	0.903
	±0.13%	±0.05%	±8.98%	±4.83%	±22.22%	±6.09%	±0.159	±0.042
Sand	99.95%	99.94%	74.17%	90.00%	85.00%	60.00%	0.791	0.713
	±0.04%	±0.04%	±1.95%	±31.82%	±15.91%	±0.01%	±0.005	±0.119
Plants	99.88%	99.87%	94.09%	96.30%	95.92%	93.32%	0.949	0.943
	±0.05%	±0.05%	±3.86%	±3.04%	±5.93%	±3.06%	±0.024	±0.021

misidentification of metal as wood. This misclassification is likely rooted in the similar smooth textures both materials exhibit, causing difficulty distinguishing them accurately. Please take a look at [Table 6](#) for more details.

4.5. Language-based zero-shot classification

Our method has bridged the visual and tactile modalities; we want to delve deeper into the discrepancies between the generated data and the original data in terms of language alignment performance. So, we employed a language-based zero-shot classification task. Using the earlier trained visual-tactile model, we extract visual and tactile representations. Next, we calculate the similarity between them and the text representation in the language model, respectively, selecting the text with the highest similarity as the predicted category. Although our tactile model was not co-trained with the verbal model, as outlined in ImageBind ([Girdhar et al., 2023](#)), it benefits from the alignment of tactile and visual modalities. This alignment implicitly ensures a connection between the tactile and verbal modalities. The experimental results show that the alignment of the generated tactile data to the language representation is very close to that of the original data, as the classification performance is roughly comparable. However, the diminished classification performance of the generated visual data suggests a reduced consistency with the language representation. This supports our prior claim that generated tactile data are superior to visual ones. Refer to [Table 7](#) for detailed results.

We initially examine the cosine distance between language embeddings spanning different categories within the latent feature space. In the language latent feature space, synthetic and natural fibers are

strikingly similar, having a mere 0.06 cosine distance. This closeness causes classification confusion between these categories, impacting performance. This issue manifested conspicuously in the original data and generated images during our analysis. In tactile, many synthetic fibers are mislabeled as natural fibers, while natural fibers are typically classified correctly.

As before, we computed the F1 scores for language-based classification of generated versus original data, as depicted in [Fig. 8](#). We noted that the F1 scores for the generated visual data consistently lagged behind those of the original data across various categories. The rubber category was notably impacted, dropping from 0.056 to 0.011, implying that most of the generated rubber images were misclassified. Only two actual rubber images were correctly labeled, hinting at an under-representation of rubber in the dataset. The sand also stood out with a 51.43% F1 score drop, from 0.042 to 0.020. The decline was attributed to misclassifications of synthetic fiber and concrete as sand. This indicates the under-representation of Syn-Fiber and concrete in the generated dataset.

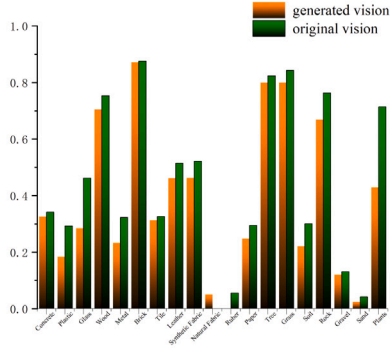
We performed a comparable analysis for generated tactile data. Neither generated nor actual rubber data were classified correctly for tactile data. Given the visual results, this might stem from a scarcity of rubber data. Like the overall accuracy, F1 scores for generated and actual tactile data were consistent across categories. This suggests our generated tactile data closely match the quality of the original tactile data.

Our experiments addressed the three initial questions. For the first question, we validated our method on two datasets, showcasing strong generalization. For the second, using various metrics, generated data of our method proved highly competitive. For the last, we performed

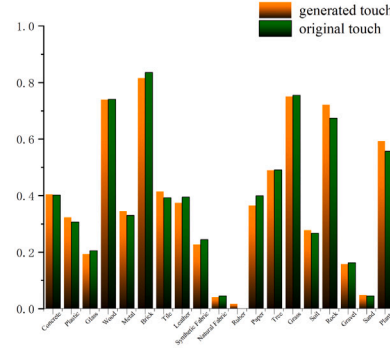
Table 7

Results on zero-shot classification, for generated data we report the means and 95% confidence intervals calculated over 4 seeds, for original data we only report results from one experiment, since we did not retrain the CLIP model.

	method	Accuracy↑	Precision↑	Recall↑	F1 score↑
language-visual	original data	50.33%	52.57%	49.61%	0.510
	generated data	44.49%	43.67%	42.64%	0.432
		$\pm 0.58\%$	$\pm 1.10\%$	$\pm 1.26\%$	± 0.011
language-tactile	original data	44.92%	59.48%	40.01%	0.479
	generated data	45.03%	41.10%	59.06%	0.485
		$\pm 0.43\%$	$\pm 0.51\%$	$\pm 1.64\%$	± 0.007



(a) The F1 score of generated vision and original vision.



(b) The F1 score of generated touch and original touch.

Fig. 8. The zero-shot classification of the language-visual-tactile alignment model, the F1 scores in our generated, and original data with means and 95% confidence intervals calculated over 4 seeds.

language-based zero-shot classification experiments, demonstrating the data generated can be applied to other tasks. Our approach demonstrated generation capabilities across various categories without notable deficiencies in any class, albeit with potential for further improvement.

5. Discussion and conclusion

Inspired by the internal information processing of the human brain, we present a visual-tactile cross-modal generative algorithm facilitating bidirectional mapping in latent feature spaces. This is the first study to leverage a flow model approach for bidirectional mapping between vision and tactile sensations. We improve the coherence between visual and tactile modalities by utilizing priors to simplify the mapping process and reduce computational demands by the VAE model. Furthermore, our exploration into the influence of various priors revealed that category information in our algorithm can shape image style, with the semantic code dictating flow direction, enhancing the transparency of our model.

We evaluated our approach on two datasets, each with distinct tactile representations. Using various metric dimensions for evaluation, our approach consistently surpassed existing methods. Notably, our performance in zero-shot language classification tasks indicates that our generated data are apt for downstream task learning—a boon for applications facing data collection challenges. Our approach offers promising support for space robot teleoperation by enabling synchronous generation of pseudo-tactile data from visual observations, thereby enriching human–robot interaction diversity.

As our experiments show, the limitation in generating visual signals for intricate textures of our method requires further refinement. Upon examining the consistency between our generated visual and generated tactile, we observed a lesser alignment than original data an issue we need to address in the future. Our subsequent focus will be on multi-modal information generation encompassing vision, tactile, and language. This approach aims for a broader generation across multiple modalities, a pivotal step towards achieving genuine intelligence.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 62373129, the National Natural Science Foundation of China (No. 92048301), the National Natural Science Foundation of China (No. U21B6002), and the Heilongjiang Provincial Natural Science Foundation of China (YQ2023F012).

References

- Ardizzone, L., Kruse, J., Lüth, C., Bracher, N., Rother, C., & Köthe, U. (2021). Conditional invertible neural networks for diverse image-to-image translation. In *Pattern recognition: 42nd DAGM german conference, DAGM GCPR 2020, Tübingen, Germany, September 28–October 1, 2020, proceedings 42* (pp. 373–387). Springer.
- Cai, S., Zhu, K., Ban, Y., & Narumi, T. (2021). Visual-tactile cross-modal data generation using residue-fusion gan with feature-matching and perceptual losses. *IEEE Robotics and Automation Letters*, 6(4), 7525–7532.
- Chu, V., McMahon, I., Riano, L., McDonald, C. G., He, Q., Perez-Tejada, J. M., et al. (2015). Robotic learning of haptic adjectives through physical interaction. *Robotics and Autonomous Systems*, 63, 279–292.
- Crick, F., & Koch, C. (1990). Towards a neurobiological theory of consciousness. In *Seminars in the neurosciences, vol. 2* (pp. 263–275). Saunders Scientific Publications.
- Cui, S., Wang, R., Wei, J., Hu, J., & Wang, S. (2020). Self-attention based visual-tactile fusion learning for predicting grasp outcomes. *IEEE Robotics and Automation Letters*, 5(4), 5827–5834.
- Gao, Y., Hendricks, L. A., Kuchenbecker, K. J., & Darrell, T. (2016). Deep learning for tactile understanding from visual and haptic data. In *2016 IEEE international conference on robotics and automation* (pp. 536–543). IEEE.

- Girdhar, R., El-Nouby, A., Liu, Z., Singh, M., Alwala, K. V., Joulin, A., et al. (2023). Imagebind: One embedding space to bind them all. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 15180–15190).
- Isola, P., Zhu, J.-Y., Zhou, T., & Efros, A. A. (2017). Image-to-image translation with conditional adversarial networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 1125–1134).
- Lee, J.-T., Bollegala, D., & Luo, S. (2019). “Touching to see” and “seeing to feel”: Robotic cross-modal sensory data generation for visual-tactile perception. In *2019 International conference on robotics and automation* (pp. 4276–4282). IEEE.
- Li, M., Sareh, S., Xu, G., Ridzuan, M. B., Luo, S., Xie, J., et al. (2016). Evaluation of pseudo-haptic interactions with soft objects in virtual environments. *PLoS One*, 11(6), Article e0157681.
- Li, Y., Zhu, J.-Y., Tedrake, R., & Torralba, A. (2019). Connecting touch and vision via cross-modal prediction. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 10609–10618).
- Liu, X., Gong, C., & Liu, Q. (2022). Flow straight and fast: Learning to generate and transfer data with rectified flow. arXiv preprint arXiv:2209.03003.
- Liu, H., Guo, D., Zhang, X., Zhu, W., Fang, B., & Sun, F. (2020). Toward image-to-tactile cross-modal perception for visually impaired people. *IEEE Transactions on Automation Science and Engineering*, 18(2), 521–529.
- Liu, H., Wang, F., Sun, F., & Fang, B. (2018). Surface material retrieval using weakly paired cross-modal learning. *IEEE Transactions on Automation Science and Engineering*, 16(2), 781–791.
- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., et al. (2021). Learning transferable visual models from natural language supervision. In *International conference on machine learning* (pp. 8748–8763). PMLR.
- Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. (2022). High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 10684–10695).
- Saharia, C., Chan, W., Chang, H., Lee, C., Ho, J., Salimans, T., et al. (2022). Palette: Image-to-image diffusion models. In *ACM SIGGRAPH 2022 conference proceedings* (pp. 1–10).
- Sanchez, G., Hartmann, T., Fuscà, M., Demarchi, G., & Weisz, N. (2020). Decoding across sensory modalities reveals common supramodal signatures of conscious perception. *Proceedings of the National Academy of Sciences*, 117(13), 7437–7446.
- Sann, C., & Streri, A. (2007). Perception of object shape and texture in human newborns: evidence from cross-modal transfer tasks. *Developmental Science*, 10(3), 399–410.
- Senellart, A., Chadebec, C., & Allasconnière, S. (2023). Improving multimodal joint variational autoencoders through normalizing flows and correlation analysis.
- Seymour, K., Clifford, C. W., Logothetis, N. K., & Bartels, A. (2009). The coding of color, motion, and their conjunction in the human visual cortex. *Current Biology*, 19(3), 177–183.
- Simonyan, K., & Zisserman, A. (2015). Very deep convolutional networks for large-scale image recognition. In *International conference on learning representations, international conference on learning representations*.
- Smith, F. W., & Goodale, M. A. (2015). Decoding visual object categories in early somatosensory cortex. *Cerebral Cortex*, 25(4), 1020–1031.
- Strese, M., Schuwerk, C., Iepure, A., & Steinbach, E. (2016). Multimodal feature-based surface material classification. *IEEE Transactions on Haptics*, 10(2), 226–239.
- Suzuki, M., & Matsuo, Y. (2022). A survey of multimodal deep generative models. *Advanced Robotics*, 36(5–6), 261–278.
- Suzuki, M., Nakayama, K., & Matsuo, Y. (2016). Joint multimodal learning with deep generative models. *Learning, Learning*.
- Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., & Wojna, Z. (2016). Rethinking the inception architecture for computer vision. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 2818–2826).
- Ujitoko, Y., & Ban, Y. (2018). Vibrotactile signal generation from texture images or attributes using generative adversarial network. In *Haptics: science, technology, and applications: 11th international conference, EuroHaptics 2018, Pisa, Italy, June 13–16, 2018, proceedings, Part II 11* (pp. 25–36). Springer.
- Xu, W., Yu, Z., Xue, H., Ye, R., Yao, S., & Lu, C. (2023). Visual-tactile sensing for in-hand object reconstruction. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 8803–8812).
- Yang, F., Ma, C., Zhang, J., Zhu, J., Yuan, W., & Owens, A. (2022). Touch and go: Learning from human-collected vision and touch. arXiv preprint arXiv:2211.12498.
- Yu, Z., Xu, W., Ren, J., Tang, T., Li, Y., Yao, S., et al. (2023). Precise robotic needle-threading with tactile perception and reinforcement learning. In *7th Annual conference on robot learning*.
- Yuan, W., Dong, S., & Adelson, E. H. (2017). Gelsight: High-resolution robot tactile sensors for estimating geometry and force. *Sensors*, 17(12), 2762.
- Yuan, W., Wang, S., Dong, S., & Adelson, E. (2017). Connecting look and feel: Associating the visual and tactile properties of physical materials. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 5580–5588).
- Zhang, R., Isola, P., Efros, A. A., Shechtman, E., & Wang, O. (2018). The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 586–595).
- Zhao, Y., Xie, J., & Li, P. (2020). Learning energy-based generative models via coarse-to-fine expanding and sampling. In *International conference on learning representations*.
- Zhu, J.-Y., Park, T., Isola, P., & Efros, A. A. (2017). Unpaired image-to-image translation using cycle-consistent adversarial networks. In *Proceedings of the IEEE international conference on computer vision* (pp. 2223–2232).
- Zhu, J.-Y., Zhang, R., Pathak, D., Darrell, T., Efros, A. A., Wang, O., et al. (2017). Toward multimodal image-to-image translation. *Advances in Neural Information Processing Systems*, 30.